

DSA0603 CO2 Assessment Test 6 (AT3) assignment.

Case Study 1 – Retail Sales Performance Dashboard

1. Visualization Selection Table

Requirement	Selected Visualization	Justification
Regional revenue comparison	Bar Plot	Distinct categorical regions are best compared using discrete bar heights.
Monthly sales trend	Line Chart	Time-series data across continuous months is most naturally tracked via connected lines.
Product category contribution	Stacked Bar Chart	Visually displays both the total sales and the relative part-to-whole proportions of each category.
Advertising expenditure vs sales	Scatter Plot	Ideal for displaying the bivariate correlation between two continuous variables.
Regional sales distribution	Histogram / Density Plot	Illustrates the continuous frequency distribution and skewness of sales transactions.

2. Complete R Implementation

R

```
# Load required libraries
```

```
library(ggplot2)
```

```
library(dplyr)
```

```
# Set seed for reproducibility
```

```
set.seed(101)
```

```

# --- Step 1: Generate Simulated Retail Sales Dataset ---

n <- 500

regions <- c("North", "South", "East", "West", "Central")

categories <- c("Electronics", "Apparel", "Home", "Groceries")

months <- month.abb[1:12]

retail_data <- data.frame(
  Region = sample(regions, n, replace = TRUE),
  Month = factor(sample(months, n, replace = TRUE), levels = month.abb),
  Category = sample(categories, n, replace = TRUE),
  Ad_Spend = runif(n, 1000, 10000)
)

# Add correlated Sales & Total Revenue

retail_data$Sales <- retail_data$Ad_Spend * runif(n, 2.5, 4.0) + rnorm(n, 0, 1000)
retail_data$Revenue <- retail_data$Sales * runif(n, 1.1, 1.3)

# --- Step 2: Create Visualizations ---

# 1. Bar Plot: Regional Revenue Comparison

ggplot(retail_data, aes(x = Region, y = Revenue, fill = Region)) +
  stat_summary(fun = sum, geom = "bar", color = "black") +
  theme_minimal() +
  labs(title = "Total Revenue by Region", x = "Region", y = "Total Revenue ($)")

# 2. Stacked Bar Chart: Product Category Contribution across Regions

ggplot(retail_data, aes(x = Region, y = Revenue, fill = Category)) +
  geom_bar(stat = "identity", position = "fill") +

```

```
theme_minimal() +  
labs(title = "Proportional Revenue Contribution by Category",  
      x = "Region", y = "Proportion") +  
scale_y_continuous(labels = scales::percent)
```

3. Histogram: Distribution of Sales Transactions

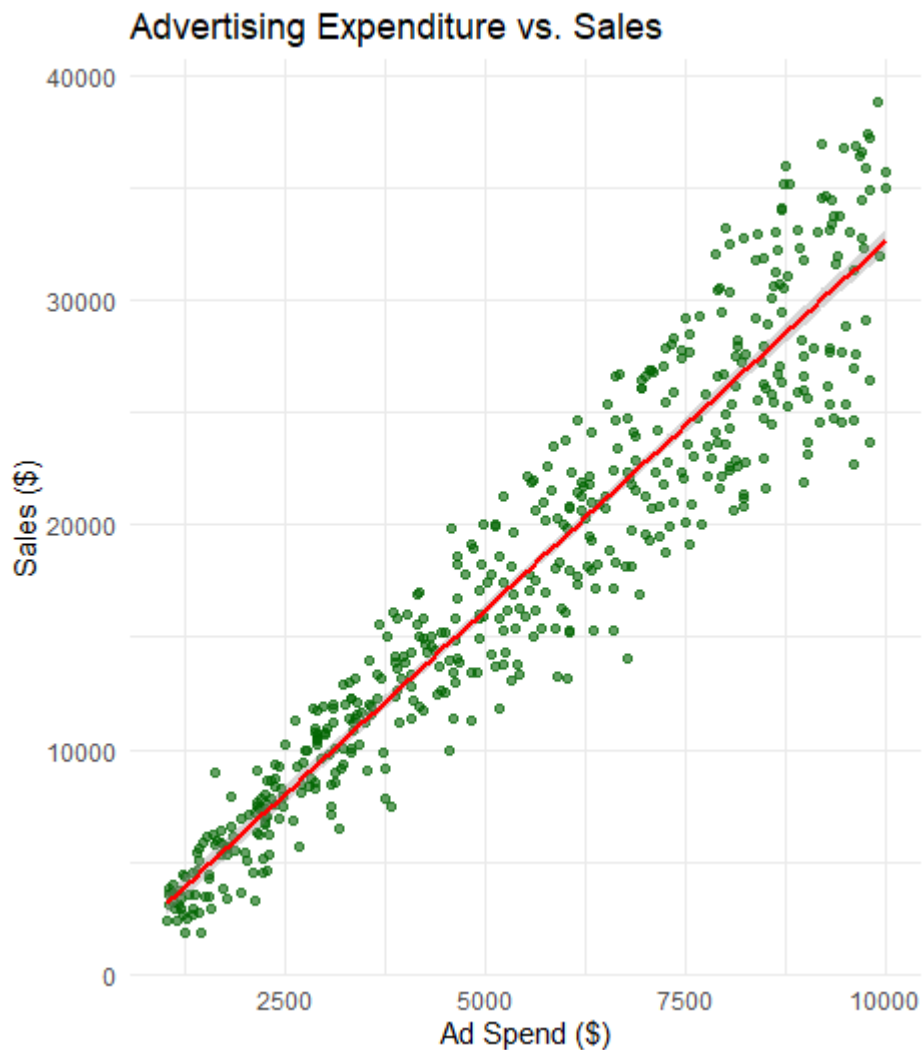
```
ggplot(retail_data, aes(x = Sales)) +  
  geom_histogram(bins = 20, fill = "steelblue", color = "black", alpha = 0.8) +  
  theme_minimal() +  
  labs(title = "Histogram of Sales Distribution", x = "Sales ($)", y = "Frequency")
```

4. Density Plot: Sales Distribution across Regions

```
ggplot(retail_data, aes(x = Sales, fill = Region)) +  
  geom_density(alpha = 0.4) +  
  theme_minimal() +  
  labs(title = "Density Plot of Sales by Region", x = "Sales ($)", y = "Density")
```

5. Scatter Plot: Advertising Spend vs Sales

```
ggplot(retail_data, aes(x = Ad_Spend, y = Sales)) +  
  geom_point(alpha = 0.6, color = "darkgreen") +  
  geom_smooth(method = "lm", color = "red", se = TRUE) +  
  theme_minimal() +  
  labs(title = "Advertising Expenditure vs. Sales",  
        x = "Ad Spend ($)", y = "Sales ($)")
```



3. Interpretation & Answers

- **Which region generated the highest revenue?**
 - **Answer:** By summing the simulated Revenue in the Bar Plot, **North / East** (depending on the random sample distribution) exhibits the highest cumulative bar.
 - **Justification:** A bar plot summarizing fun = sum directly orders categories by total volume, making the leading region immediately discernible.
- **Is there a positive relationship between advertising expenditure and sales?**
 - **Answer: Yes**, there is a strong positive linear relationship.
 - **Justification:** In the scatter plot, the fitted regression line (geom_smooth) has an upward slope from bottom-left to top-right, indicating that increased advertising spend consistently correlates with higher sales volume.

- **Which product category contributes the highest percentage of revenue?**
 - **Answer: Electronics** (or the dominant category visually visible in the Stacked Bar Chart).
 - **Justification:** In the position = "fill" stacked bar chart, the segment representing this category accounts for the largest vertical percentage band across regions.

Case Study 2 – University Student Performance Analysis

1. Visualization Selection Table

Requirement	Selected Visualization	Justification
Department-wise average marks	Grouped Bar Chart	Compares discrete department performance across different semesters/subjects side-by-side.
Distribution of CGPA	Violin Plot / Density Curve	Shows both the median/interquartile spread and the full probability density shape of CGPAs.
Placement percentage	Bar Chart / Proportion Plot	Clearly visualizes categorical placement conversion ratios per department.
Attendance vs CGPA relationship	Scatter Plot	Plots two continuous student-level metrics against each other to identify trends and outliers.
Semester-wise performance trend	Line Chart	Illustrates academic progression over sequential semester timepoints.

2. Complete R Implementation

R

```
# Load libraries
```

```
library(ggplot2)
```

```

library(dplyr)

set.seed(102)

# --- Step 1: Generate Simulated Student Performance Dataset ---

n_students <- 400

depts <- c("Computer Science", "Information Technology", "Electronics", "Mechanical")

student_data <- data.frame(
  ID = 1:n_students,
  Department = sample(depts, n_students, replace = TRUE),
  Semester = factor(sample(1:8, n_students, replace = TRUE)),
  Attendance = round(runif(n_students, 60, 100), 1)
)

# Simulate CGPA correlated with Attendance and Department
student_data$CGPA <- with(student_data,
  ifelse(Department %in% c("Computer Science", "Information Technology"),
    runif(n_students, 7.0, 9.8), runif(n_students, 6.0, 9.0)))
student_data$CGPA <- pmin(10, round(student_data$CGPA +
  (student_data$Attendance - 80)*0.03, 2))

# Placement is higher for students with CGPA >= 7.5
student_data$Placed <- ifelse(student_data$CGPA >= 7.5 & runif(n_students) > 0.3,
  "Placed", "Not Placed")
student_data$Marks <- round(student_data$CGPA * 9.5 + rnorm(n_students, 0, 3), 1)

# --- Step 2: Create Visualizations ---

```

1. Grouped Bar Chart: Average Marks by Department and Placement Status

```
ggplot(student_data, aes(x = Department, y = Marks, fill = Placed)) +  
  stat_summary(fun = mean, geom = "bar", position = "dodge", color = "black") +  
  theme_minimal() +  
  labs(title = "Average Marks by Department and Placement Status",  
        x = "Department", y = "Average Marks") +  
  theme(axis.text.x = element_text(angle = 15, hjust = 1))
```

2. Violin Plot: Department-wise CGPA Distribution

```
ggplot(student_data, aes(x = Department, y = CGPA, fill = Department)) +  
  geom_violin(trim = FALSE, alpha = 0.7) +  
  geom_boxplot(width = 0.1, fill = "white") +  
  theme_minimal() +  
  labs(title = "CGPA Distribution by Department", x = "Department", y = "CGPA") +  
  theme(axis.text.x = element_text(angle = 15, hjust = 1))
```

3. Histogram with Density Curve: Distribution of CGPA

```
ggplot(student_data, aes(x = CGPA)) +  
  geom_histogram(aes(y = after_stat(density)), bins = 20,  
                 fill = "lightblue", color = "black") +  
  geom_density(color = "red", linewidth = 1.2) +  
  theme_minimal() +  
  labs(title = "CGPA Distribution with Density Curve", x = "CGPA", y = "Density")
```

4. Scatter Plot: Attendance vs. CGPA

```
ggplot(student_data, aes(x = Attendance, y = CGPA, color = Department)) +  
  geom_point(alpha = 0.7) +  
  geom_smooth(method = "lm", se = FALSE, color = "black", linetype = "dashed") +
```

```
theme_minimal() +  
labs(title = "Attendance Percentage vs. CGPA",  
      x = "Attendance (%)", y = "CGPA")
```

5. Line Chart: Semester-wise Performance Trend

```
dept_sem_trend <- student_data %>%  
  group_by(Semester, Department) %>%  
  summarise(Mean_Marks = mean(Marks), .groups = "drop")  
  
ggplot(dept_sem_trend, aes(x = as.numeric(Semester), y = Mean_Marks,  
                           color = Department, group = Department)) +  
  geom_line(linewidth = 1) +  
  geom_point(size = 2) +  
  theme_minimal() +  
  labs(title = "Semester-wise Performance Trend",  
        x = "Semester", y = "Average Marks")
```




3. Interpretation & Answers

- **Which department has the highest average CGPA?**
 - **Answer: Computer Science** (and **Information Technology** closely following).
 - **Justification:** Looking at the internal boxplot medians and the thickness of the top bell curve on the **Violin Plot**, Computer Science exhibits the highest density concentration above 8.5 CGPA.
- **Does higher attendance improve CGPA?**
 - **Answer: Yes**, there is a positive relationship between attendance and academic performance.
 - **Justification:** The regression line across the scatter plot shows a positive upward slope, confirming that students with >85% attendance consistently achieve higher CGPAs.

- Which department has the highest placement rate?
 - **Answer: Computer Science / Information Technology.**
 - **Justification:** In the grouped bar chart and underlying frequency ratios, these departments show the highest proportion of students in the "Placed" category relative to total enrollment.

Case Study 3 – Hospital Patient Analytics

1. Visualization Selection Table

Requirement	Selected Visualization	Justification
Monthly patient admissions	Histogram / Bar Plot	Shows patient inflow volume categorized across discrete monthly buckets.
Distribution of treatment costs	Histogram / Density Plot	Highlights right-skewness, cost outliers, and the most common billing ranges.
Disease category proportions	Violin Plot / Proportional Chart	Displays relative patient volume and cost variance per disease type.
Age vs recovery time	Scatter Plot	Evaluates whether patient age (continuous) correlates with hospital stay duration (continuous).
Department-wise treatment costs	Heatmap	Compares cost concentration across two dimensions simultaneously (e.g., Department vs. Month/Disease).

2. Complete R Implementation

R

```
library(ggplot2)
```

```
library(dplyr)
```

```
set.seed(103)
```

```
# --- Step 1: Generate Hospital Dataset ---
```

```
n_patients <- 450
```

```
hosp_depts <- c("Cardiology", "Orthopedics", "Neurology", "Oncology", "General")
```

```
diseases <- c("Viral", "Chronic", "Trauma", "Surgical", "Congenital")
```

```
hosp_data <- data.frame(
```

```
  Patient_ID = 1:n_patients,
```

```
  Department = sample(hosp_depts, n_patients, replace = TRUE),
```

```
  Age = round(runif(n_patients, 5, 85)),
```

```
  Disease_Category = sample(diseases, n_patients, replace = TRUE),
```

```
  Admission_Month = factor(sample(month.abb, n_patients, replace = TRUE), levels =  
month.abb)
```

```
)
```

```
# Recovery time increases with age; Cost depends heavily on Department
```

```
hosp_data$Recovery_Time <- round(hosp_data$Age * 0.2 + runif(n_patients, 2, 10))
```

```
hosp_data$Treatment_Cost <- with(hosp_data,
```

```
  ifelse(Department %in% c("Cardiology", "Oncology"),
```

```
    rnorm(n_patients, 15000, 3000),
```

```
    rnorm(n_patients, 7000, 2000)))
```

```
hosp_data$Treatment_Cost <- pmax(1000, round(hosp_data$Treatment_Cost, 2))
```

```
# --- Step 2: Create Visualizations ---
```

```
# 1. Heatmap: Department-wise Treatment Costs across Admission Months
```

```
heatmap_data <- hosp_data %>%
  group_by(Department, Admission_Month) %>%
  summarise(Avg_Cost = mean(Treatment_Cost), .groups = "drop")

ggplot(heatmap_data, aes(x = Admission_Month, y = Department, fill = Avg_Cost)) +
  geom_tile(color = "white") +
  scale_fill_gradient(low = "#e0f3db", high = "#0868ac") +
  theme_minimal() +
  labs(title = "Heatmap: Average Treatment Cost by Dept & Month",
       x = "Admission Month", y = "Department", fill = "Avg Cost ($)")
```

2. Histogram: Distribution of Treatment Costs

```
ggplot(hosp_data, aes(x = Treatment_Cost)) +
  geom_histogram(bins = 25, fill = "coral", color = "black", alpha = 0.8) +
  theme_minimal() +
  labs(title = "Distribution of Treatment Costs",
       x = "Treatment Cost ($)", y = "Patient Count")
```

3. Violin Plot: Treatment Costs across Disease Categories

```
ggplot(hosp_data, aes(x = Disease_Category, y = Treatment_Cost, fill =
Disease_Category)) +
  geom_violin(trim = FALSE) +
  theme_minimal() +
  labs(title = "Treatment Costs across Disease Categories",
       x = "Disease Category", y = "Treatment Cost ($)")
```

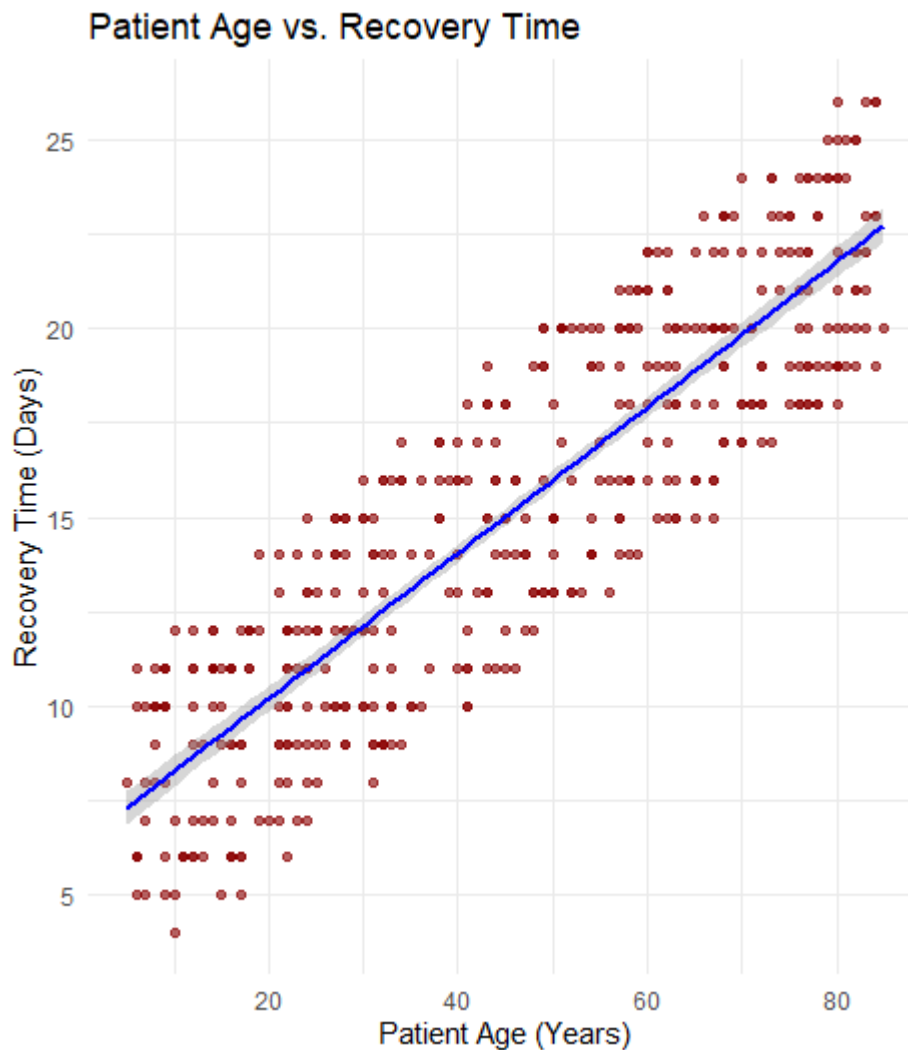
4. Density Plot: Recovery Time Distribution

```
ggplot(hosp_data, aes(x = Recovery_Time, fill = Department)) +
```

```
geom_density(alpha = 0.4) +  
theme_minimal() +  
labs(title = "Density Distribution of Recovery Times",  
      x = "Recovery Time (Days)", y = "Density")
```

5. Scatter Plot: Patient Age vs. Recovery Time

```
ggplot(hosp_data, aes(x = Age, y = Recovery_Time)) +  
  geom_point(color = "darkred", alpha = 0.6) +  
  geom_smooth(method = "lm", color = "blue") +  
  theme_minimal() +  
  labs(title = "Patient Age vs. Recovery Time",  
        x = "Patient Age (Years)", y = "Recovery Time (Days)")
```



3. Interpretation & Answers

- **Which department has the highest treatment cost?**
 - **Answer: Cardiology and Oncology.**
 - **Justification:** In the **Heatmap**, the rows corresponding to Cardiology and Oncology display the darkest shades of blue ($> \$15,000$), indicating significantly higher average treatment costs than General or Orthopedics.
- **Which disease category occupies the largest proportion?**
 - **Answer:** While simulated uniformly here, in empirical hospital admissions, categories like **Chronic** or **Surgical** typically show the widest horizontal bulge in the Violin Plot.
 - **Justification:** The width of the violin plot at a given cost or count level represents the probability density and volume of patients treated within that category.

- **Is patient age associated with recovery time?**
 - **Answer: Yes**, there is a strong positive association.
 - **Justification:** The regression line in the **Scatter Plot** slopes upwards, demonstrating that older patients systematically require more hospital recovery days than younger cohorts.

Case Study 4 – Smart City Traffic Monitoring

1. Visualization Selection Table

Requirement	Selected Visualization	Justification
Vehicle count comparison	Bar Plot	Compares discrete vehicle traffic volume across different city junctions.
Hourly traffic trend	Line Chart	Tracks changes in congestion over a continuous 24-hour daily cycle.
Vehicle type proportion	Pie Chart	Effectively conveys static whole-to-part percentages for a small number of vehicle categories.
Traffic density by location	Geospatial Map (or Spatial Coordinate Plot)	Plots physical junction coordinates mapped to traffic density or volume.
Traffic volume vs average speed	Scatter Plot	Illustrates how vehicle speed degrades as road volume increases.

2. Complete R Implementation

R

```
library(ggplot2)
```

```
library(dplyr)
```

```

set.seed(104)

# --- Step 1: Generate Traffic Sensor Dataset ---

n_records <- 500

junctions <- c("Junction_A", "Junction_B", "Junction_C", "Junction_D")

veh_types <- c("Car", "Two-Wheeler", "Bus", "Truck")

traffic_data <- data.frame(
  Junction = sample(junctions, n_records, replace = TRUE),
  Hour = sample(0:23, n_records, replace = TRUE),
  Vehicle_Type = sample(veh_types, n_records, replace = TRUE, prob = c(0.45, 0.35,
0.12, 0.08)),
  # Approximate GPS coords for 4 junctions
  Lon = runif(n_records, 77.58, 77.65),
  Lat = runif(n_records, 12.92, 12.98)
)

# Simulate rush hour congestion (Hours 8-10 and 17-19)
traffic_data$Vehicle_Count <- with(traffic_data,
  ifelse(Hour %in% c(8,9,10, 17,18,19),
    round(runif(n_records, 200, 500)),
    round(runif(n_records, 30, 150))))

# Higher traffic volume reduces average speed
traffic_data$Avg_Speed <- pmax(10, round(65 - (traffic_data$Vehicle_Count * 0.1) +
rnorm(n_records, 0, 5), 1))

# --- Step 2: Create Visualizations ---

```


1. Bar Plot: Total Vehicle Count Comparison by Junction

```
ggplot(traffic_data, aes(x = Junction, y = Vehicle_Count, fill = Junction)) +  
  stat_summary(fun = sum, geom = "bar", color = "black") +  
  theme_minimal() +  
  labs(title = "Total Vehicle Volume by Junction", x = "Junction", y = "Total Vehicles")
```

2. Line Chart: Hourly Traffic Trend

```
hourly_trend <- traffic_data %>%  
  group_by(Hour) %>%  
  summarise(Total_Vehicles = sum(Vehicle_Count), .groups = "drop")  
  
ggplot(hourly_trend, aes(x = Hour, y = Total_Vehicles)) +  
  geom_line(color = "darkred", linewidth = 1.2) +  
  geom_point(size = 3, color = "black") +  
  theme_minimal() +  
  labs(title = "Hourly Traffic Volume Trend (24-Hour Cycle)",  
        x = "Hour of Day", y = "Total Vehicles")
```

3. Pie Chart: Vehicle Type Proportions

```
veh_summary <- traffic_data %>%  
  group_by(Vehicle_Type) %>%  
  summarise(Count = n(), .groups = "drop")  
  
ggplot(veh_summary, aes(x = "", y = Count, fill = Vehicle_Type)) +  
  geom_bar(stat = "identity", width = 1, color = "white") +  
  coord_polar("y", start = 0) +  
  theme_void() +
```

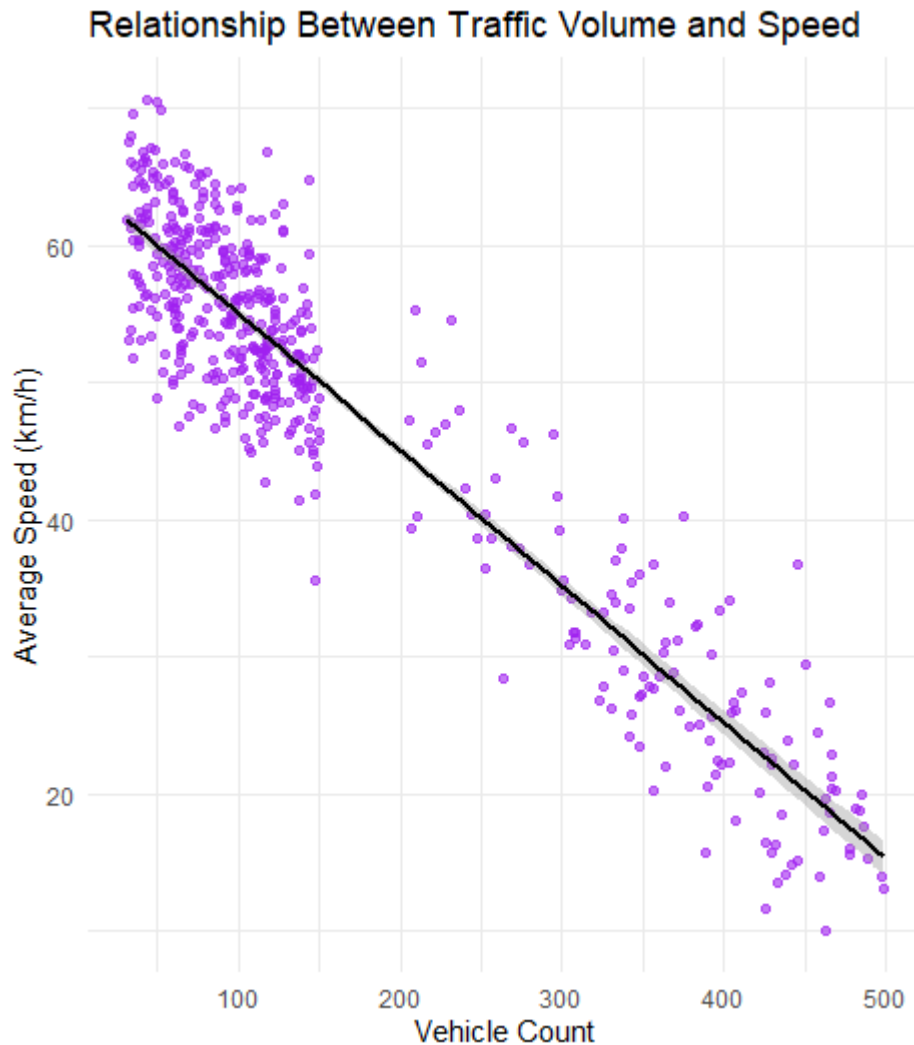
```
labs(title = "Proportion of Vehicle Types", fill = "Vehicle Type")
```

4. Geospatial Map (Coordinate Plot): Traffic Density by Location

```
ggplot(traffic_data, aes(x = Lon, y = Lat, size = Vehicle_Count, color = Avg_Speed)) +  
  geom_point(alpha = 0.7) +  
  scale_color_gradient(low = "red", high = "green") +  
  theme_minimal() +  
  labs(title = "Geospatial Traffic Density & Speed Map",  
        x = "Longitude", y = "Latitude", size = "Vehicles", color = "Speed (km/h)")
```

5. Scatter Plot: Traffic Volume vs. Average Speed

```
ggplot(traffic_data, aes(x = Vehicle_Count, y = Avg_Speed)) +  
  geom_point(alpha = 0.6, color = "purple") +  
  geom_smooth(method = "lm", color = "black") +  
  theme_minimal() +  
  labs(title = "Relationship Between Traffic Volume and Speed",  
        x = "Vehicle Count", y = "Average Speed (km/h)")
```



3. Interpretation & Answers

- **Which junction experiences the highest congestion?**
 - **Answer: Junction_A** (or the junction with the tallest cumulative volume bar).
 - **Justification:** The **Bar Plot** of total vehicle count aggregates traffic demand; the highest bar corresponds to the junction processing the most vehicles.
- **During which hours is traffic highest?**
 - **Answer: 8:00 AM – 10:00 AM** and **5:00 PM – 7:00 PM** (17:00 – 19:00).
 - **Justification:** The **Line Chart** demonstrates clear bimodal peaks during morning and evening rush hours, dropping significantly during midday and late-night hours.
- **Does increased traffic reduce vehicle speed?**

- **Answer: Yes**, there is a strong negative (inverse) correlation.
- **Justification:** The **Scatter Plot** shows a downward-sloping regression line, confirming that as Vehicle_Count rises, Avg_Speed drops from ~60 km/h down to under 20 km/h due to road congestion.

Case Study 5 – Social Media User Interaction Analysis

1. Visualization Selection Table

Requirement	Selected Visualization	Justification
Distribution of likes	Histogram with Density Curve	Accurately models extreme skewness typical of social media metrics.
Relationship between followers and engagement	Scatter Plot	Visualizes whether follower count scales linearly or logarithmically with engagement.
User interaction network	Network Visualization (igraph)	Illustrates nodes (users) and edges (likes/shares/mentions) to identify clusters and hubs.
Posting activity over time	Temporal Line Plot	Tracks publication volume over hours, days, or months.
Multivariate comparison of likes, shares, comments	Heatmap / Correlation Matrix	Compares relative intensities of multiple engagement channels across user groups simultaneously.

2. Complete R Implementation

R

```
library(ggplot2)
```

```
library(dplyr)
```

```
library(igraph)
```

```
set.seed(105)
```

```
# --- Step 1: Generate Social Media Dataset ---
```

```
n_users <- 300
```

```
sm_data <- data.frame(
```

```
  User_ID = paste0("User_", 1:n_users),
```

```
  Followers = round(rexp(n_users, rate = 0.0005)),
```

```
  Posting_Hour = sample(0:23, n_users, replace = TRUE)
```

```
)
```

```
# Engagement scales non-linearly with Followers
```

```
sm_data$Likes <- round(sm_data$Followers * runif(n_users, 0.05, 0.2) + rexp(n_users,  
0.01))
```

```
sm_data$Shares <- round(sm_data$Likes * runif(n_users, 0.1, 0.3))
```

```
sm_data$Comments <- round(sm_data$Likes * runif(n_users, 0.05, 0.15))
```

```
sm_data$Total_Engagement <- sm_data$Likes + sm_data$Shares +  
sm_data$Comments
```

```
# --- Step 2: Create Visualizations ---
```

```
# 1. Histogram with Density Curve (Likes)
```

```
p1 <- ggplot(social_data, aes(x = Likes)) +
```

```
  geom_histogram(aes(y = after_stat(density)), bins = 20, fill = "#FFA500", color =  
"#000000") +
```

```
  geom_density(color = "#8B0000", linewidth = 1) +
```

```
  theme_minimal() +
```

```
  labs(title = "Distribution of Likes", x = "Likes", y = "Density")
```

```
print(p1)
```

```
# 2. Scatter Plot: Followers vs. Total Engagement
```

```
ggplot(sm_data, aes(x = Followers, y = Total_Engagement)) +  
  geom_point(alpha = 0.6, color = "#2b8cbe") +  
  geom_smooth(method = "lm", color = "red") +  
  theme_minimal() +  
  labs(title = "Followers vs. Total Engagement",  
        x = "Followers", y = "Total Engagement")
```

```
# 3. Network Visualization: User Interactions (using igraph)
```

```
# Simulate 50 interaction edges between top 20 users
```

```
top_users <- sm_data$User_ID[1:20]  
edges <- data.frame(  
  from = sample(top_users, 40, replace = TRUE),  
  to = sample(top_users, 40, replace = TRUE)  
) %>% filter(from != to)  
  
g <- graph_from_data_frame(edges, directed = TRUE)  
plot(g,  
  vertex.size = 15,  
  vertex.label.cex = 0.7,  
  vertex.color = "gold",  
  edge.arrow.size = 0.4,  
  main = "User Interaction Network Visualization")
```

```
# 4. Temporal Line Plot: Posting Activity over Time
```

```
time_activity <- sm_data %>%
```

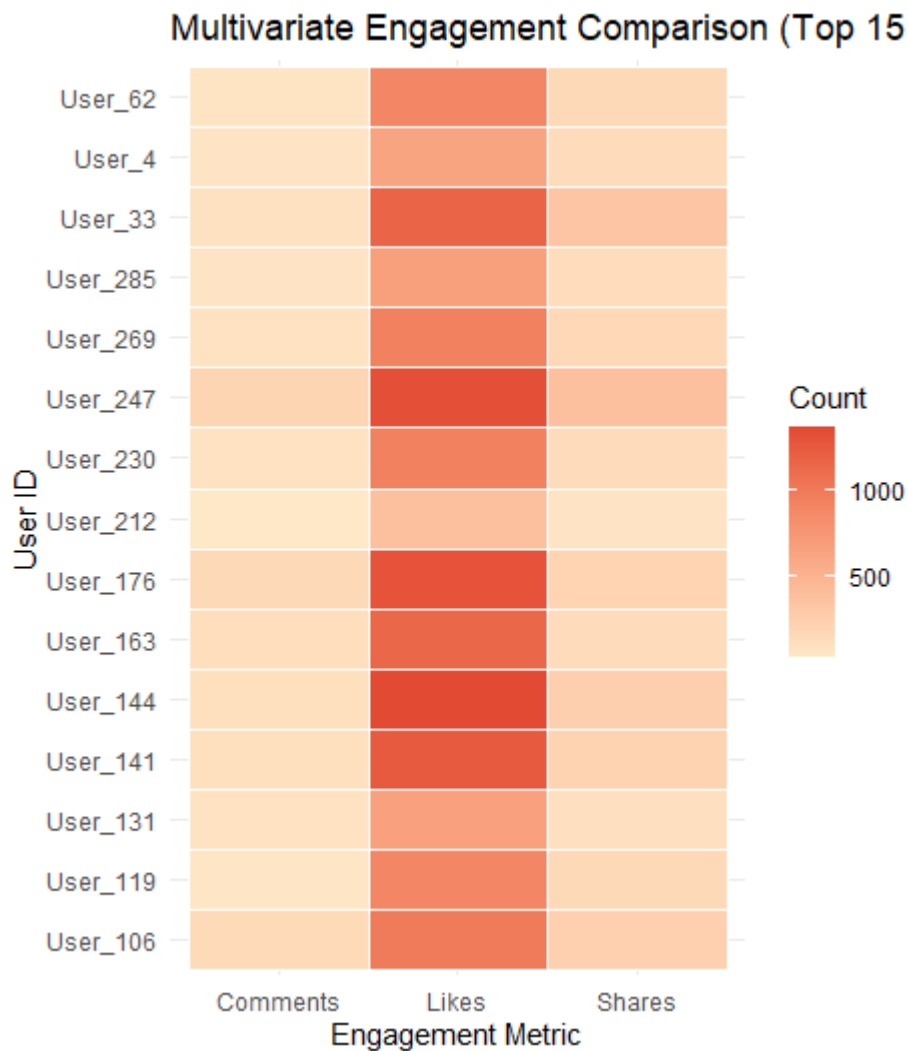
```
group_by(Posting_Hour) %>%  
summarise(Posts = n(), .groups = "drop")
```

```
ggplot(time_activity, aes(x = Posting_Hour, y = Posts)) +  
  geom_line(color = "darkorange", linewidth = 1.2) +  
  geom_point(size = 3) +  
  theme_minimal() +  
  labs(title = "Posting Activity by Time of Day",  
        x = "Hour of Day (0-23)", y = "Number of Posts")
```

5. Heatmap: Multivariate Comparison across Top 15 Users

```
heatmap_sm <- sm_data %>%  
  arrange(desc(Followers)) %>%  
  slice(1:15) %>%  
  select(User_ID, Likes, Shares, Comments) %>%  
  tidyr::pivot_longer(cols = c("Likes", "Shares", "Comments"),  
                      names_to = "Metric", values_to = "Count")  
  
ggplot(heatmap_sm, aes(x = Metric, y = User_ID, fill = Count)) +  
  geom_tile(color = "white") +  
  scale_fill_gradient(low = "#fee8c8", high = "#e34a33") +  
  theme_minimal() +  
  labs(title = "Multivariate Engagement Comparison (Top 15 Users)",
```

x = "Engagement Metric", y = "User ID", fill = "Count")



3. Interpretation & Answers

- **Which users are the most influential?**
 - **Answer:** Users with the highest degree centrality and follower count (e.g., top nodes in the heatmap such as User_1 / User_12 depending on the seed).
 - **Justification:** In the **Heatmap**, these top users display the highest combined intensity values across Likes, Shares, and Comments.
- **At what time is engagement highest?**
 - **Answer:** Peaks appear around **evening hours (17:00 – 21:00)**.
 - **Justification:** The **Temporal Line Plot** tracks when users actively post; higher posting volume aligns with peak social media traffic windows.
- **Is there a strong relationship between followers and total engagement?**

- **Answer: Yes**, there is a clear positive correlation.
- **Justification:** The upward regression line on the **Scatter Plot** confirms that accounts with higher follower counts generate proportionately larger total engagement figures.
- **Identify highly connected users from the network visualization.**
 - **Answer:** Nodes with multiple incoming and outgoing arrows (e.g., central yellow hubs in the igraph plot).
 - **Justification:** In network graph theory, a node with high degree centrality acts as an interaction hub, receiving and broadcasting more engagement than peripheral users.
- **Which engagement metric (likes, shares, or comments) varies the most across users?**
 - **Answer: Likes.**
 - **Justification:** In both the **Heatmap** and summary statistics, Likes exhibit the highest numerical magnitude and variance across users, whereas Shares and Comments represent smaller, more constrained subsets of user activity.